Journal of Nondestructive Evaluation (2024) 43:14
https://doi.org/10.1007/s10921-023-01032-x

Advanced Faster-RCNN Model for Automated Recognition and Detection of Weld Defects on Limited X-Ray Image Dataset

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Received: 4 October 2023 / Accepted: 21 November 2023 / Published online: 16 December 2023
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Abstract
Computer-aided weld defect recognition is transforming the field of Non-Destructive Testing by addressing the shortcomings of slow and error-prone manual inspections. This technology provides a reliable solution for detecting changes in pipeline conditions and structural damage. While conventional neural networks fall short in precise fault localization, deep learning-based object detection techniques step in to fill the gap. Addressing a real-industrial problem, particularly visually inspecting an X-ray welding database, without relying on a pre-existing benchmark presents a significant challenge in this field. Additionally, the poor quality of our welding data, which is riddled with small, sticky porosity in each image, poses several issues related to selecting the appropriate deep neural network object detector. This is yet another challenge that needs to be tackled. To direct these challenges, we introduced a novel approach based on the renowned Faster RCNN architecture to develop a model specifically designed for weld defect detection and recognition. This study dives deep into the inner workings of this newly adopted methodology. In our research, we have thoroughly parameterized, trained, tested, and validated this model. Our approach stands out through a comparative analysis with YOLO and DCNN models, highlighting the superiority of our Faster RCNN-based system. By evaluating its robustness and efficiency, our study reveals that the Faster RCNN model outperforms its counterparts in weld defect detection and localization for this specific small and sticky porosity defect type. This stands as a testament to effectively setting a new standard in this area.

Keywords Weld defect detection · X-ray images · Object detection · Faster RCNN

1 Introduction

In the process of constructing water pipes, both internal and external welding operations are performed to secure the metal parts together. These critical welding tasks are subject to scrutiny by human experts who possess the ability to identify various types of welding defects, including cracks and porosities, which can arise from imperfections in the joints. This fact could lead to the limited life of the pipes. So, quality control is necessary to guarantee the best quality of the pipelines. The piping check must be carried out without destroying the component. Online nondestructive testing (NDT) is currently tested using machine vision techniques. Industries employ inspection techniques to verify that the required specifications of a material, structure, or system are being met, all while ensuring the integrity of the original part remains intact. Within this field, welds undergo inspection through NDT techniques like radiography, wherein the radiation is passed through the specimen to identify defects. X-ray imaging is particularly effective for inspecting thin materials, of offering detailed visibility and resolution to detect defects with precision. On the other hand, gamma-ray imaging is employed when dealing with thicker components. To facilitate analysis and processing, the results can be digitized using film radiography, computed radiography, or computed tomography. However, digitized images often exhibit low quality and diminished contrast, making the examination of weld defects a laborious task. Unfortunately, these characteristics interfere with object interpretation, classification, and recognition. As a result, artificial intelligence technology for computer vision applications is invented to

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help the expert evaluate the results. In this context, these methods are essential research approaches that replace the skilled professional in weld defect detecting and localizing, offering a neutral and cost-effective alternative. To achieve a complete comprehension of the image, our objective extends beyond image categorization. We strive to precisely estimate the identities and precise positions of objects present within each image. This methodology is commonly referred to as object detection [1], encompassing various sub-tasks such as face detection [2], pedestrian detection [3], and spinal detection [4]. To gain a complete understanding of the image, we do not only focus on sorting the different images but also seek to accurately approximate the concepts and locations of the contained objects in each image. This approach is called object detection [1], which generally includes different sub-tasks such as face detection [2], pedestrian detection [3] and spinal detection [4]. Object detection is a crucial field in computer vision, offering valuable insights for comprehending images and videos. Its significance is evident in various applications such as image classification [5], human behavior analysis [6], facial recognition [7], and autonomous driving [8]. The advancement of neural networks and related learning systems has greatly influenced the development of network algorithms and revolutionized object recognition techniques. However, the challenges posed by diverse viewpoints, occlusions, poses, and lighting conditions make achieving precise object recognition with accurate localization a complex task. As a result, this area has garnered substantial attention in recent years from researchers and practitioners alike [9–11]. Object detection involves two primary tasks: determining the spatial position of objects within a frame (object location) and assigning the appropriate class label to each object (object classification). With the advent of deep neural networks (DNN) [12] in the field, significant advancements have been made, particularly with the introduction of region-based convolutional neural networks (R-CNN) [9]. These innovations have brought about substantial improvements in the detection and recognition of objects, enabling more accurate and efficient object localization and classification. Their deeper architectures enable the learning of complicated features compared to shallow ones. The strength of expressiveness and the robust learning of algorithms also allow the learning representations of informative objects without having to manually design features [13].

Many upgraded Models have been suggested as RCNN proposals, which jointly optimize classification tasks and bounding box regression [9]. Fast RCNN utilizes another sub-grid to generate regional proposals, allowing for precise object localization and classification [10]. On the other hand, YOLO (You Only Look Once) performs object recognition using a fixed grid regression approach, enabling real-time object detection and making it suitable for consumer products such as Google Photos and Pinterest Visual Search [11]. These methods have demonstrated fast and efficient performance, even on mobile devices. Moreover, the availability of pre-trained models with varying degrees of recognition performance improvement has made it easier to achieve accurate object recognition and real-time rendering. However, professionals may find it challenging to determine the best architecture for their specific application. Standard metrics like mean average precision (mAP) provide limited insight, as other factors such as computational implementation, execution time, and memory consumption also play crucial roles in determining the suitability of a vision system.

In fact, this problem of precision and accuracy in the welding area is due, on the one hand, to the variety of classes of weld defects [14], on the other hand, to the fact that the characteristics generated are insufficient for obtaining a good recognition quality in order to effectively detect the fault. Classic machine learning (ML) algorithms have been widely researched and applied, following the traditional ML application scheme and employing feature mining techniques. Despite the time-consuming and less accurate nature of certain ML algorithms like Artificial Neural Networks (ANN), their ease of implementation has been recognized. However, with the emergence of deep learning, the utilization of large datasets and automatic feature extraction has significantly improved defect detection capabilities. In the same context, there are a number of studies aimed at optimizing such a control method with the objective of creating an automatic X-ray image inspection and analysis system, but the application of Deep Learning techniques in weld defect detection remains challenging with limited prior work or established methods in this area.

Research has been conducted in the literature on deep learning models for weld defect detection based on pre-trained convolutional neural network models [15]. Nevertheless, adopting a dedicated deep learning architecture for weld defect detection to suit specific data continues to pose a significant challenge that could be approached from various angles. In order to overcome the limitations of previous work and address the complexities of our dataset, we leverage Transfer Learning by applying a pre-trained network [16].

Our objective is to customize a network model tailored to our specific use case and to detect multiple defects of new data sets with respectable precision. As the usual trade of the object detection process is applied for the RGB images and generally separated objects, the detection algorithm becomes more granted. On the contrary, the new trade of X-ray datasets consisting of sticky defects with small sizes makes the detection procedure more complex.

Recently, works based on deep learning have done breakthroughs in perception-based applications but, a big amount of data is always needed to form a model from scratch. For

instance, Faster RCNN [10] demands around 150,000 natural images to obtain the highest yield and lack of training data poses problems in using this new technology.

The insufficiency of X-ray welding images is a challenge that needs to be adjusted especially if they are acquired with low quality and a variety of types. Moreover, the detection of X-ray weld defects with the abundance of small sticky porosity in every single image leads to several issues related to the selection of the proper deep neural network object detector and its performance. For these reasons, this technology based on a deep learning algorithm to detect defects in welds is newly done. In this paper, we demonstrate that by using a very small training data set, high accuracy, and higher efficiency can be achieved by developing a deep learning network previously trained on natural images. So, we adjust the Faster RCNN network using 453 X-ray welding images and we compare it with traditional and new models of detection. In this article we seek to explore execution time and testing results and why not the performance at the time of testing. Within this study, comprehensive research is carried out to outline the AI models and diverse properties across multiple application areas together with the welding field. The content of this article is organized as follows. Section 2 gives a brief introduction to the deep learning background and the fundamental structure of CNN. In Sect. 3, the architecture and structure of object detection are presented and detailed 3. Then, the Faster RCNN application results are shown and compared to the other models in Sects. 4–6, respectively. At last, some concluding remarks are presented and further potential directions in the future are outlined in Sect. 7.

2 Review of Existing Studies

Object detection is a significant area of research within the computer vision sector. Traditional methods heavily relied on handcrafted features like the histogram of oriented gradients (HOG) and Scale-invariant feature transformation (SIFT), along with multiple classifiers. However, the performance of these traditional approaches falls short when compared to the recent breakthroughs in deep learning [12, 17, 18].

In recent years, neural networks have emerged as the primary method for object detection, revolutionizing the field. This section highlights some notable contributions to the literature. One of the pioneering works in detection based on convolutional networks was the R-CNN article by Girshick et al. [19].

Building upon the advancements of R-CNN, Fast R-CNN [9] introduced an improved approach to address computational inefficiencies. Both R-CNN and Fast R-CNN relied on external proposal generators for bounding box suggestions. However, recent developments in the field have demonstrated the potential of utilizing neural networks for generating box proposals as well [10, 20–23]. This advancement eliminates the need for separate proposal generation, streamlining the object detection pipeline and further enhancing the efficiency and accuracy of the overall system.

Several methods have focused on accelerating the detection process [1, 19, 24]. One notable advancement is the Faster R-CNN proposed by Ren, which employs a convolutional neural network as a Region Proposal Network (RPN) [10]. This technique significantly reduces the object detection time while achieving high accuracy [18]. However, training a network like Faster R-CNN from scratch requires a substantial amount of labeled training data. This poses challenges when applying this technology to X-ray imaging applications, as collecting a large dataset with accurately labeled images becomes extremely difficult.

In recent research, deep networks have emerged as effective tools for predicting bounding boxes of objects. In this research [23], an overFeat model is applied where a fully connected layer can predict the box coordinates for single object localization. This fully connected layer is then transformed into a convolutional layer to enable the detection of multiple class-specific objects. Building upon this, the MultiBox methods [21, 22] have introduced a novel approach where region proposals are generated from a network's last fully connected layer, simultaneously predicting independent class boxes. This ex- tends the"one box" mode of overFeat and these class-independent boxes are subsequently utilized in the R-CNN framework [19]. It is worth noting that the MultiBox proposal network can be applied to both single-image cropping and multiple cropping of large images, such as 224 × 224 dimensions, which differentiates it from our entirely convolutional scheme. Additionally, MultiBox does not share functionalities between the proposal and detection networks, offering a unique perspective on object detection techniques.

While ANN-based flaw detection has demonstrated success in various do- mains, its adoption on production lines remains limited [25]. The conventional approach involves employing mathematical morphology techniques to segment images and extract defect characteristics. For instance, Kasban et al. [26]. pro- posed a feature extraction method as a foundation for defect recognition, while Mery et al. [27]. focused on detecting weld defects based on texture attributes. Valavanis et al. [28] presented a method combining texture and geometric characteristics for detection and classification. Gang et al. [29] present a novel approach, which involves the implementation of an auto- mated classification method for weld defects. This method incorporated image processing techniques, pattern recognition methodologies, and feature extraction algorithms. The researchers applied the BSM technique to accurately bring out the defective regions from the images and employed both the fuzzy nearest neighbor algorithm K and MLP to accomplish the classification tasks

separately. Other studies utilized artificial neural networks (ANNs) for classification after feature extraction [30, 31], with Abouelatta et al.[32]. proposing a different classifier construction method based on image histogram and neural networks. Principal component analysis (PCA) in combination with ANN was used for feature reduction [33], while SVM classifiers were employed for more efficient calculations based on texture and morphological characteristics [34]. The integration of Principal Component Analysis (PCA) and Support Vector Machines (SVM) was implemented to enhance the accuracy of the system, as reported by the study [35]. Another approach put forth by Baniukiewicz involved the development of a classifier that combined an ANN with a fuzzy logic process [36]. In a different approach, Benzhi et al.[37]. developed a model for defect detection in weld radiographs using sparse reconstruction. By analyzing the regional characteristics of the bands, the researchers were able to locate the weld area and distinguish between non-cracking and cracking defects. The categorization process relied on waveform analysis of the gray column to obtain a rough classification. Although the current method relies on manual feature design, advancements are being made to mitigate these challenges.

Generally, the application of deep learning for object detection has gained significant traction in the radiography field. By incorporating the expertise and knowledge of ANN professionals, deep learning models utilize the ANN's structure, feature selection, and extraction related to the learning to achieve effective defect recognition. While transfer learning methods reduce the re- quired training data, there is still a need for increased defect data to enhance training. Recent research on the fundamental principles of CNN and related test series has paved the way for the application of transfer learning techniques in welding. In my latest publication [38], I proposed an Alex Net classification network comprising five convolutional layers and applied fine-tuning to classify welding images. This approach has demonstrated high accuracy in Welding classification tasks, showcasing the potential of deep learning in welding applications.

In my previous work, I outlined a training approach where an initial model is trained using a specific dataset. To adapt the model to X-ray data, which often exhibits a different class distribution or even introduces new classes, we further train the same model with this X-ray dataset. While training from scratch can yield superior results, it comes at a higher cost in terms of time and resource consumption. Alternatively, leveraging a pre-trained model in a related task proves highly effective, particularly through fine-tuning. By selectively freezing pre-trained layers and only training the added layers, we achieve significant performance gains in X-ray data analysis. This approach strikes a balance between leveraging prior knowledge and adapting to the unique characteristics of the X-ray dataset, ultimately leading to enhanced model performance.

When adjusting the learning rate, it is important to consider the impact on the training process. Increasing the learning rate too much can result in poor outcomes as the gradient descent optimization takes large steps. This can cause the neural network to converge to a local minimum instead of the de- sired global minimum. On the other hand, insufficient data in the new dataset or suboptimal hyperparameters can also lead to unsatisfactory results. The effectiveness of machine learning heavily relies on the dataset and network parameters. To address this, augmenting the data with techniques like translation, rotation, and flipping can be beneficial. Additionally, fine-tuning the model based on the augmented data often yields improved results. A more detailed discussion on data description is written in Sect. 4.2.1.

Convolutional Neural Networks (CNN) have demonstrated significant potential in the field of object detection. Specifically, they have been shown to be highly effective in analyzing X-ray and ultrasound images. Their capacity to analyze extensive sets of X-ray images, identify defects, and determine their respective classes stands out as a remarkable advancement in this field. An AI model has recently been applied to perform semantic segmentation of micrographs of complex weld areas to automatically detect and quantify weld seam properties [39]. A new approach to identifying weld defects has been developed using deep learning [40]. The approach involves using a Convolutional Neural Network (CNN) to extract high-level features from images of the weld surface defects, simplifying the feature extraction process. The classification is done using the Random Forest Classification algorithm, which is known for its strong generalization ability and robustness. Several studies have shown the effectiveness of CNN in detecting defects in welds. In 2018, a study conducted by the authors of [41] proposed a weld defect detection method for steel sur- faces using the Faster RCNN network, which achieved high precision across the five defect classes.

Another approach [42] in 2018 where the authors conducted a comprehensive comparison of the performances of two object detection algorithms (Faster-RCNN and SSD) in detecting and classifying defects. The study revealed that the YOLO network has limitations in detecting smaller objects.

So, when it comes to the welding domain, the application of CNNs is still a challenging task. Despite its promise, CNN still faces several limitations in this field, such as the difficulty in detecting weld defects due to the com- plex nature of welding images and the need for specialized training datasets to improve accuracy. Researchers in [43], utilize Faster R-CNN, to recognize defects in welding within the WDXI dataset. This methodology encompassed both multi-classification and precise localization techniques, enabling

accurate identification and positioning of welds in the given data. Additional study [44] is conducted to interpret radiograph test images by extracting features and classifying anomalies using an AI model. However, these methods still require a large training dataset, which is often impractical. Moreover, the training time of neural networks, a crucial factor in practical applications, was not adequately considered. In the present work, we implemented one of the deep models for object detection, Faster RCNN, to detect and localize one of the mean weld defects called porosity. We will compare our results with various combinations of pre-trained networks and with another object detection model called YOLOv2 and finally, we will validate it with various metrics.

3 Methodology

3.1 Faster RCNN Architecture

RCNN, a Convolutional Neural Network (CNN)-based object detection algorithm, employs a filter of predefined dimensions to search for objects within an image across all its regions. As illustrated in Fig. 1, Faster RCNN introduces an innovative approach by incorporating a region proposal network (RPN) into the traditional selective search method, facilitating the generation of region proposals. This neural network model enables learning through the configuration of convolutional and fully connected layers. In addition, fast computation is feasible by using graphics processor units (GPU). Intermediate layers are generated by RPN shown in Fig. 1 through a sliding window and an extractor with 256 or 512 entity dimensions. Subsequently, the convolution operation is conducted on two layers: the classification layer and the regression layer. The classifier employs a 1 × 1 filter to produce outputs. This process results in the generation of K anchor boxes, encompassing a range of sizes and ratios. For every sliding window, two scores are allocated to indicate an object's presence.

In the second layer, a 1 × 1 filter is utilized to generate k anchor boxes, followed by the allocation of four coordinate values representing the bounding box coordinates. Learning takes place by feeding the outputs from both layers into the fully connected layer via the region of interest (RoI) pooling layer. The integration of the RoI pooling layer and the region proposal network (RPN) in Faster R-CNN effectively tackles the computational and structural challenges faced by RCNN and Fast R-CNN architectures, resulting in significantly improved accuracy and calculation speed.

With a faster RCNN model, errors are reduced thanks to the RPN module that propagates backward from one end to the other. Given the convolutional basis of the RPN structure, the feature extractor can be also coupled to it. Accordingly, the efficiency can be enhanced and high results on Image-net are yielded by implementing multiple CNN structures.

3.2 RPN: Region Proposal Networks

The region proposal network (RPN) takes the input data and generates a set of object proposals, assigning an objectness score[1] to each proposal. This model is built upon a fully convolutional network, which allows for the simultaneous prediction of object boundaries and scores at every position. To capture this process, we employ a fully convolutional network [45] as outlined in this section.

The RPN structure, depicted in Fig. 1, operates by sliding across the convolutional feature map and fully connecting to a spatial window with dimensions n × n. This operation generates a low-dimensional vector (512-d for VGG16, 256-d for ZF) for each sliding window, which is then fed into two fully connected (FC) layers: the box regression layer (reg) and the box classification layer (cls). To implement this model, an n × n convolutional layer and two 1 × 1 convolutional layers are utilized. Additionally, ReLU activation is applied to enhance non-linearity in the output of the n × n convolutional layer. The regressions are performed to assess the proposals using anchors (refer to Sect. 3.2.1).

Specifically, anchors with 3 scales and 3 proportions are considered. The loss function remains the same as in the Eq. 1.

$$L(p_i, t_i) = \frac{1}{N_{cls}} \sum_i L_{cls}((p_i, p_i^*)) + \lambda \frac{1}{N_{reg}} \sum_i \left(p_i^* L_{reg}\left(t_i, t_i^*\right)\right) \quad (1)$$

p_i defines the predicted probability of the i-th anchor as an object, where p^* is equal to 1 if the anchor is positive and 0 otherwise. t_i saves the predicted bounding box four coordinates while t^* corresponds to the coordinates of the ground truth box covered by a positive anchor. The loss function L_{cls} is is a logarithmic binary loss and L_{reg} is a smoothed L1 loss the same as cited [9] and illustrated below 2:

$$\text{Smooth } L1(x) = \begin{matrix} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{matrix}$$

$p_i^* L_{reg}$ This term is defined when the regression loss is only enabled for positive anchors $\left(p_i^* = 1\right)$ and is deactivated otherwise $\left(p_i^* = 0\right)$—p_i and t_i are the cls and reg layers outputs respectively. These latter modules are regularized to (N_{cls}) and (N_{reg}) respectively and weighted by an equilibrium

[1] "Objectness" estimates membership in a group of object classes with respect to the background.

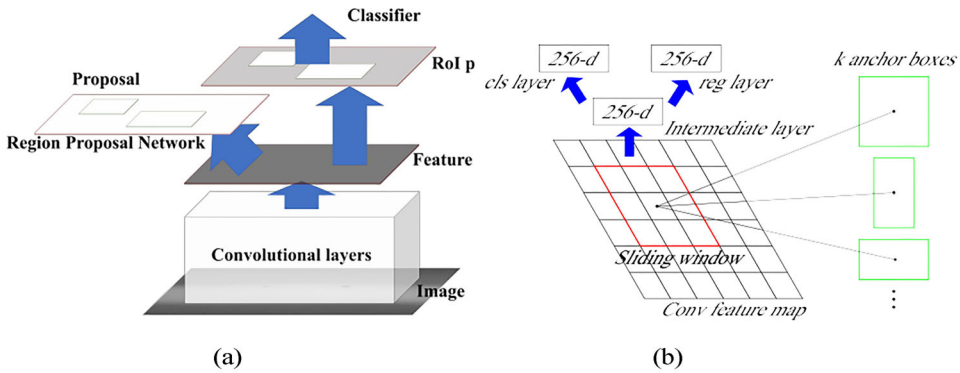

Fig. 1 **a** Faster RCNN network for object detection: it consists of two models, one module is a fully convolutional deep region proposal network, and another module is the fast RCNN detector. **b** RPN module that generates object proposals from an image and defines an objectiveness score for each one [10]

parameter λ. In our running implementation, the cls term in Eq. 1 is standardized by the mini-batch size, and the term reg is standardized by the anchor locations number. As λ = 10 by default, the cls and reg terms have approximately similar weights. Faster RCNNs may be formed by an end-to-end model using backpropagation and SGD on the different forming matter.

3.2.1 Anchor Concept

Multiple region statements for every sliding window position are predicted at one time. The most probable number region in each location is specified by k. The cls layer serves the purpose of generating 2 k scores, which are used to predict the presence or absence of an object. Meanwhile, the reg layer produces 4 k outputs, representing the coordinates of K boxes. These K proposals are structured in alignment with reference boxes known as anchor boxes, which are characterized by their scale and aspect ratio (as depicted in Fig. 1, right). Typically, three scales and aspect ratios are employed, resulting in a total of K = 9 anchors for each sliding position. For example, if the convolutional feature map measures W × H in size, there would be a sum of WH_k anchors. This implies that for every region proposal, K regions are generated, each potentially varying in scale or aspect ratio.

Objectness score

The cls layer is responsible for assigning a two-element vector to each region proposal. If the first element is 1 and the second element is 0, the region is classified as background, while in the opposite case, it represents an object. The objectivity score for each anchor in the RPN is determined based on the intersection over union (IoU) metric. This metric calculates the ratio of the intersection area between the Anchorage Box and the Ground Truth Box to the area of their union. The IoU ranges from 0.0 to 1.0, where 0.0 indicates no intersection and 1.0 indicates a perfect overlap between the two boxes. There are four main conditions that determine the sign of the objectivity score.

1. An anchor is assigned a positive objectivity label if its IoU overlaps with any ground truth square greater than 0.7.
2. In case of any anchor with IoU greater than 0.7, the anchors that have the highest IoU receive a positive label.
3. Anchors with IoU overlaps less than 0.3 are assigned a negative objectivity score, indicating that they represent the background.
4. Anchors that do not fall into the positive or negative categories do not contribute to the training objective.

Note: an anchor may be considered positive even if it doesn't meet the first condition (IoU > 0.7) if no region with IoU equal to 0.7 exists, preferring the second condition (0.5 < IoU < = 0.7).

$$Objectness_{score}(IoU) = \begin{cases} Positive \rightarrow IoU > 0.7 \\ Positive \rightarrow 0.5 < IoU \leq 0.7 \\ Negative \rightarrow IoU < 0.3 \\ Not\ Negative/Positive \rightarrow 0.3 \leq IoU \leq 0.5 \end{cases}$$

3.3 Fast R-CNN Architecture

In the context of Fast R-CNN, the input comprises an image and a collection of proposed objects. The process is similar to the Region Proposal Network (RPN), with the distinction that the entire image is processed by a convolutional network (specifically, ResNet), resulting in a 512-dimensional feature map. Each object proposal is then mapped to a Region of Interest (ROI) within this feature map. After the feature extraction process, an extra layer called the ROI pooling layer is introduced to standardize the ROI features to a consistent spatial size of 14 × 14. This resultant feature vector is then fed into a fully connected layer, tasked with generating two separate output layers: one for bounding box regression and the other for probability estimation using softmax.

The softmax output provides probability estimates for different classes, including a background class. The bounding box regression output specifies the coordinates for object classes in the form of bounding box regression results. To guide the learning process for both classification and bounding box regression, a multi-task loss function, denoted as L, is defined as shown in Eq. 1.

3.4 RPN Training Strategy

It's a fully convolutional network designed for end-to-end training using back- propagation and stochastic gradient descent [46]. In this context, we follow a general training approach similar to what's described in [9], with a slight modification involving the selection of a different pre-trained model and parameter adjustments. Each image contains numerous positive and negative anchor examples, forming the basis of each mini-batch. While optimizing the loss functions for all anchors is feasible, it can introduce a bias towards negative samples, which are more dominant. As an alternative, we randomly select 256 anchors from an image to create a mini-batch, maintaining a 1:1 ratio between positive and negative anchors in the sample. If an image has fewer than 128 positive samples, we fill the mini-batch with negative samples.

Next, all new layers are initialized with random weights drawn from a Gaussian distribution with a mean of zero and a standard deviation of 0.01. Additionally, the shared convolutional layers are initialized using a Deep Residual Learning architecture previously trained for ImageNet classification [47]. In our approach, we adopt a learning rate of 0.001 for the mini-batches to accommodate the limited data, coupled with a weight decay of 0.0005 and a momentum of 0.9. For more details, refer to Sect. 5.1. Our implementation is conducted using MATLAB R 2019b,'GeForce RTX 2080', and is built upon a feature extraction network followed by two sub-networks, as explained in greater detail in Sect. 4.

3.5 Feature Exchange Between RPN and Fast R-CNN

So far, we explained the RPN sub-network and its training technique to finally generate regional proposals that will be used after by fast R-CNN model. The next steps are mainly dedicated to the layers to share between these two models. The training of each model separately leads to certain modifications in the convolutional layers. So, we tend to apply a method to join these two models and get the advantage of sharing convolutional layers between them. This technique increases learned features and thus achieves higher detection results. Each of these layers is used in both networks even if it just exists only once. This can be called layer sharing or feature sharing. Also, these are the anchors that enable the sharing of features/layers between the Faster R- CNN modules. In general, there are three ways to form networks with shared features (Approximate Joint Training, Non-Approximate Joint Training, and Alternative Training) described in [10]

The last approach "Alternative Training" begins with the formation of RPNs and takes the proposals from these regions to train Fast R-CNN. The RPN model is initialized based on the Fast R-CNN defined network and this entire sequence is repeated. So, firstly the RPN is trained as described in Sect. 3.4. This sub-network is initiated with ImageNet pre-trained model and after that is fine-tuned end to end. Secondly, Fast R-CNN is also trained on the proposals produced by the RPN model. This detection network's initialization is also done with the ImageNet model. Yet, there is no sharing of layers. So, this latter network is therefore used to initialize the learning of the RPN by defining the shared convolution layers and only fine-tuning the individual RPN layers. Thus, the layer-sharing process is done. Ultimately, the individual Fast-R-CNN layers are fine-tuned, while leaving the shared convolution layers unchanged. A comparable training can be done for other iterations, but we found insignificant improvements.

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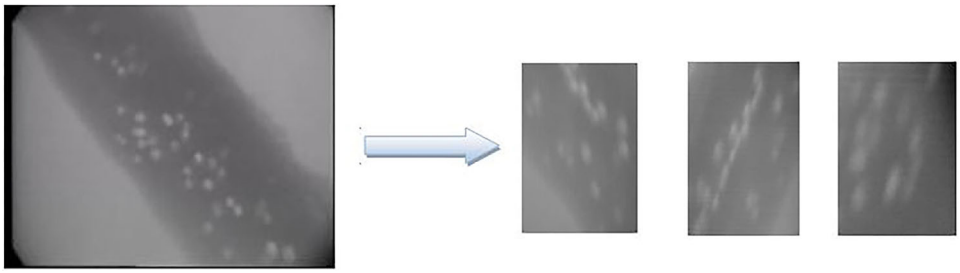


Fig. 2 Welding X-ray images resized and cropped with porosity defect type

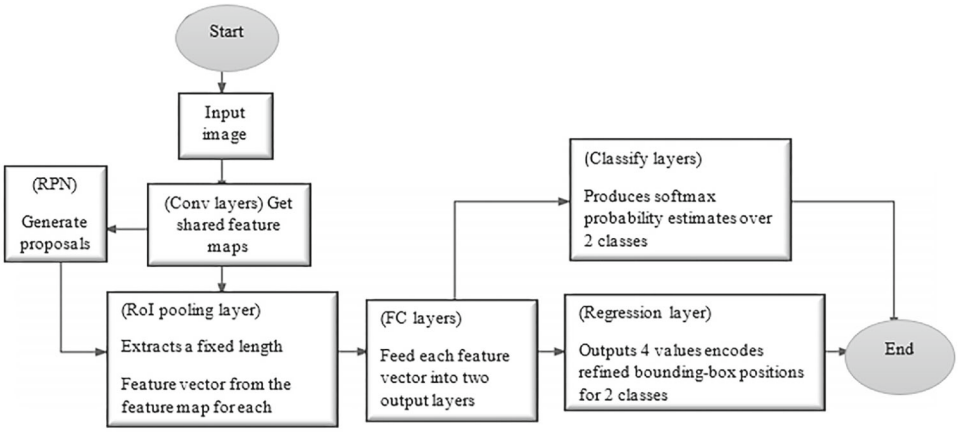


Fig. 3 Faster RCNN flowchart

layers and fine-tuning only the individual RPN layers. Consequently, the layer-sharing process takes place. Finally, the individual Fast R-CNN layers are fine-tuned, while the shared convolution layers remain unchanged. Similar training iterations can be carried out, but improvements are negligible observed.

4 Methodology Details

4.1 Challenge Overview

The X-ray images that constitute the data used in this work are produced by this system presented in Fig. 2. In addition, these data are quite small, with high-dimensional images of 640 × 480 and 720 × 576 pixels and with bad quality and uneven illumination. Each image shows one or more weld defects of varying sizes and quantities as cited in my thesis [48]. This makes weld defect detection a difficult and time-consuming task. The usual classification process uses extraction methods that have proven effective in different object recognition tasks. Given this, the birth of deep learning considerably simplifies this phase by automating the learning and extraction of features using a dedicated network structure. The major problems limiting the deep learning techniques adoption are data availability and computational power.

Given the labor-intensive nature of collecting and maintaining large enough data, general efforts on this matter have been directed to a ready data set. The main problems are the lack of databases for X-ray welds and a large number of small sticky defect types presented in every image. Moreover, the available datasets in the market (e.g. GD X-ray) [49] are not similar to our data which makes the process more difficult. For this reason, we tried to create more data from the original data samples, as shown in Fig. 2. In the present work, we will cover one major defect type, called "porosity". The images used in this study are cropped and resized from the original data that is detailed in Sect. 4.2 by keeping the same aspect ratio to not damage the quality of the image. Nevertheless, overfitting problems could result due to the insufficiency of the dataset. In order to avoid these issues and enlarge the training and testing samples, data augmentation algorithms are applied. Overall, the general steps are well explained in the next sections, and the flowchart is illustrated in Fig. 3.

4.2 Image Acquisition System

Data collection involved the utilization of an X-ray imaging system. The digitization process was carried out using various systems, such as flat detectors or a digital imaging device. Our acquisition system specifically employed the Y-Smart 160E 0.41.5 X-ray source operating at full power, the

X-ray object consisting of a spiral welded steel tube with varying thicknesses and diameters, and an image intensifier (II). The digital images obtained through this system had dimensions of 1280 × 720 pixels [38]. In this study, this digital device was utilized to convert the signal generated by Spa Maghreb tubes[2] into digital format. The resolution was carefully chosen to enable the detection and measurement of defects at a precision level within the hundredths of a millimeter range. This level of precision is particularly crucial in X-ray inspection compared to standard inspection scenarios.

Given that the quality of the produced images is sensitive to environmental conditions, as elaborated in my previous article [38], the resulting images required enhancement through filtering. Consequently, we developed an algorithm using MATLAB. This algorithm first cropped and resized the original image to create new images. Subsequently, it applied preprocessing techniques for denoising, including the Wiener filter and Gaussian filter, as well as contrast enhancement via stretching [50]. The generated images were scaled to a consistent parameter or a specific size of 250 × 200 pixels while maintaining the original aspect ratio [38].

4.2.1 Data Augmentation Techniques

The dataset we have at hand consists of only 453 labeled images, which is insufficient to train an AI model effectively. If we were to use the original dataset alone, the disparity between the learning and validation error would indicate an overfitting issue. However, incorporating more data samples can enhance the model's performance. To address this, we have implemented the data augmentation technique to expand the size of the dataset. This technique aids in improving the network's accuracy by randomly modifying the original data during training. Through the implementation of data augmentation techniques, we can increase the diversity of training data without needing more labeled samples. This technique can also resolve the partial replacement problem by reducing the gap between learning and validation errors.

In this part, we will discuss two techniques for enhancing image data. The first technique involves reflecting the image along the X-axis, which randomly flips both the image and its corresponding box labels horizontally. This is illustrated in Fig. 4. The augmented images are pre-processed based on the function that randomly resizes these images and their box labels as the input size dimensions, so-called scale augmentation. Also, it performs color augmentation that includes isolating a single-color channel. That is to say, the image can be simplified into one color channel representation with very simple single- color isolation such as R, G, and B. Our original image is considered an RGB image but with a grey scale. In that case, at each pixel, we have R = G = B and it is similar to the original image. So, we can take one of the three channels and trait it as a single-channel grey-scale image for faster computation. Typically, the test and validation data are fully authentic to the original data and should not be altered to allow for an unbiased assessment.

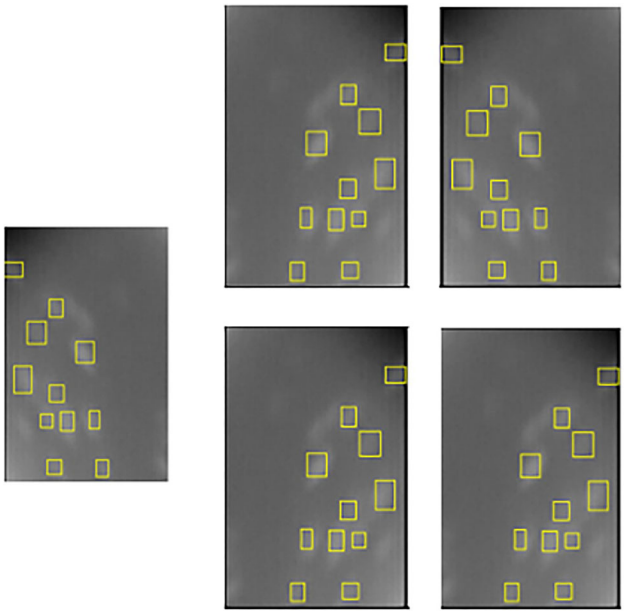


Fig. 4 An example of: original image or isolated color channel image and augmented images after reflection across the X-axis with their bboxes

4.3 AI Model Overview

4.3.1 Model Architecture

Typically, when generating a deep learning network, there's a risk of affecting the gradient's value, potentially causing saturation and, as a result, slowing down the learning process. ResNet addressed this issue by introducing a direct access connection to the identity of the conventional neural network structure. This approach enables deep network learning. The direct access connection, which links input to output without parameters, simply adds the two values together computationally. This modification significantly impacts the computational calculation. It enhances model precision by allowing for increased depth, ultimately optimizing the network. The residual learning blocks that constitute ResNet [18] are depicted in Fig. 5. This technique is then applied to facilitate learning in a direction where H(x)-x must be close to zero, making it easier to detect small input variations.

Our Faster R-CNN model consists of two key modules. The first module, known as RPN (Region Proposal Network), is responsible for generating high- quality regional proposals. These proposals serve as the foundation for the second

[2] https://smt-dz.com/

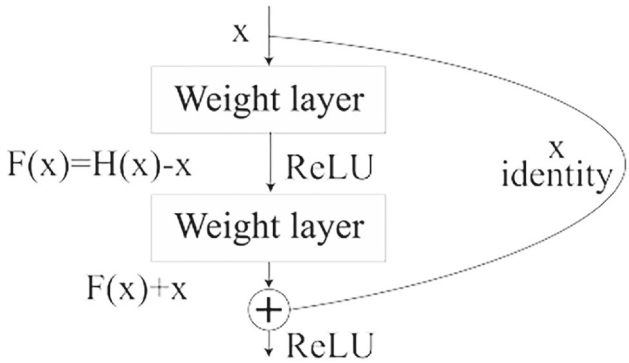


Fig. 5 Learning block of ResNet

module, Fast R-CNN, which performs detailed object detection. Essentially, the RPN module guides the Unified Unit Network on where to focus its attention. You can see the architecture in Fig. 6, which employs Fast R-CNN.

After extensive testing involving various configurations,the model described in Table 1. This chosen model comprises a total of 347 layers. To mitigate the risk of overfitting, batch normalization, and a ReLU activation layer are applied for each convolutional layer.

The RPN module incorporates a range of anchor boxes that differ in size based on the anchoring mechanism. It employs three ratios: 1:1, 1:2, and 2:1, along with three scales corresponding to box sizes of (128^2), 256^2, and 512^2 pixels. In total, there are 9 anchors at each sliding position. Increasing the image resolution in a Faster R-CNN model can enhance object recognition as it provides more information. However, considering that the dataset contains images with small defects, we've adjusted by introducing smaller anchor areas. While this adjustment aids in addressing tiny defects, it can potentially lead to longer detection processes and impact inference speed. Thus, there's a trade-off between speed and accuracy to consider.

Table 1 Faster R-CNN network

Layer name	Stride	Kernels	Kernel size	Nums
Conv_1	2	64	7 × 7	1
Max pool	2	64	3 × 3	1
Conv_2x	1	64	1 × 1	3
	1	64	3 × 3	
	1	256	1 × 1	
Conv_3x	1	128	1 × 1	4
	1	128	3 × 3	
	1	512	1 × 1	
Conv_4x	1	256	1 × 1	23
	1	256	3 × 3	
	1	1024	1 × 1	
Conv_5x	1	512	1 × 1	3
	1	512	3 × 3	
	1	2048	1 × 1	

5 Experimental Findings

This work is carried out using MATLAB. Its interface, MathWorks was used to create the deep learning framework. This framework is generated to form and train the network and perform predictions.

5.1 Faster RCNN Training Model

ResNet [18] is chosen as the feature extractor. Since our database consists of X-ray images, the ResNet model input layer is changed to work with a single- channel grey-scale image. The main database used in this work contains small data images with 453 labeled images. These input images are used with a fixed resolution equal to 224 × 224. Subsequently, this data will undergo a series of iterations, first through the convolution layer and then through the pooling activation layer, culminating in providing the results to the

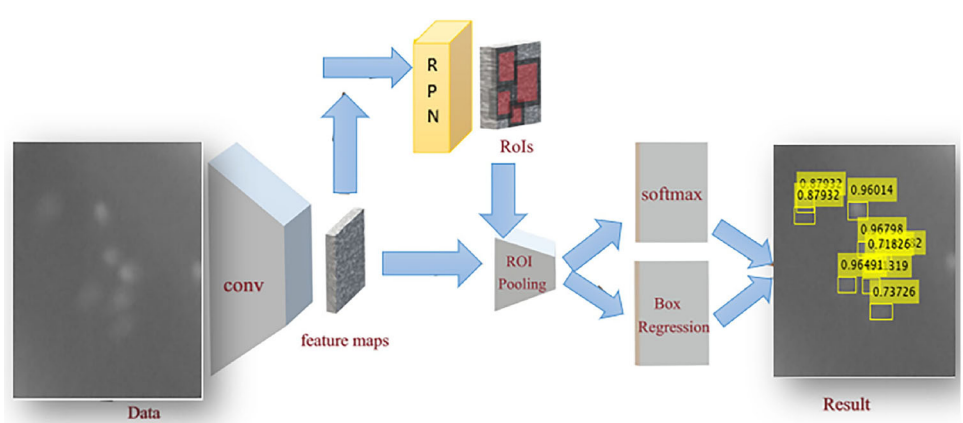


Fig. 6 Weld defect detection and recognition model structure

Table 2 Confusion matrix

	AP (%)	Mean ACC(%)	Anchors number	Mean IOU	Time (h)
Faster-RCNN	44.5%	95.15%	9	0.87	4
	31%	95%	6	0.78	4:32 min

Table 3 Performances of several models

	AP (%)	RPN ACC(%)	Mean ACC(%)	SE (%)	PRE (%)	F1 score (%)	Mean RMSE	Mean Loss
Faster Rcnn	44.5	95.15	95.26	40	63	49	0.39	0.33
ResNet50 + aug	38.1	94.5	95.01	36.7	58.58	45.17	0.55	0.41
VGG16 + aug	42.3	94.63	94.4	47	56.8	51.9	0.57	0.43
VGG19 + aug	43.4	94.12	94.38	42	60	49	0.57	0.6
AlexNet + aug	15	88.83	88.03	6	49	10	0.12	3.9
Googlenet + aug	35	93.66	94.17	39	50.7	44	0.12	0.62

fully connected layers for weld defect detection. The dataset is divided, allocating 20% for testing and reserving 80% for training purposes. As explained in Sect. 4.2.1, two different models are employed to augment the data.

The backbone network"ResNet101″ is used in the Faster RCNN model for all experiments in addition to the optimization extension of SGD algorithm with"Adam's optimizer" and a momentum of 0.9 to minimize error caused by limited data. The initial base learning rate (LR) is set to 0.001 to address this. Keeping the same learning rate helped meet the needs of various training stages. In order to reduce the error generated by the small amount of data, the initial base learning rate (LR) was set to 0.001. The main purpose of maintaining the same learning rate is to fit the requirements of the different training phases.

In general, a high learning rate is needed in the initial phase of training to accelerate it. The other parameters are adjusted as well according to the model such as the IOU being set to 0.5 and the epochs to 100. The metric of evaluating the model performance is defined by the average precision based on precision/recall. AP scores are each class's average precision (AP) scores. Typically, higher AP scores lead to improved recognition results. The algorithm-conducting machine is equipped with an Intel Core i7-9750H and NVIDIA GeForce RTX 2080, along with 32 GB of RAM. The software utilized is Windows 10 Pro, while the network model is being executed through MATLAB R 2019b.

5.2 Analysis of Outcomes and Evaluation Metrics

This subsection presents the outcomes obtained from the original Faster RCNN model and provides a comparison with the other models described in detail in the subsequent sections (see Sect. 6). To demonstrate the necessity of the localization process, we have conducted as well a comparative experiment to compare these models' performance to the original model based on ResNet- 101.So, the models are differentiated according to the feature extractor's pre- trained model name. The original model performance with different anchor values is illustrated in the following Table 2. A second table is added Table 3 to show the evaluation metrics of the different models and their descriptions are presented as follows:

$$Average\ precision\ (AP) = \int_0^1 precision(recall)\ d(recall) \tag{3}$$

$$Accuracy\ (ACC) = \frac{TP + TN}{TP + FP + FN + TN} \tag{4}$$

$$Sensitivity\ (SE) = \frac{TP}{TP + FN} \tag{5}$$

$$Precision\ (PRE) = \frac{TP}{TP + FP} \tag{6}$$

$$F1\ score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{7}$$

The first evaluation metric is Sensitivity, which refers to the rate of True prediction of porosity defect images and is called recall. The second one is Precision which refers to the percentage of correctly predicted positive images out of all positive images. So, to provide a complete evaluation of the model, it is relevant to provide as many as possible performance metrics. According to the training results, Faster RCNN has shown that using better features generated by RPN could be very beneficial. ResNet 101 is replaced by a residual network with 347 layers in this case. As a consequence, the average precision is improved and reached 44.5%

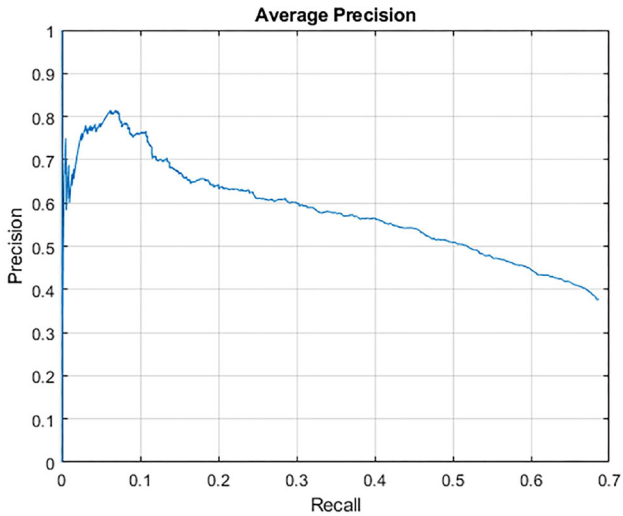


Fig. 7 Welding defect PR curve visualization

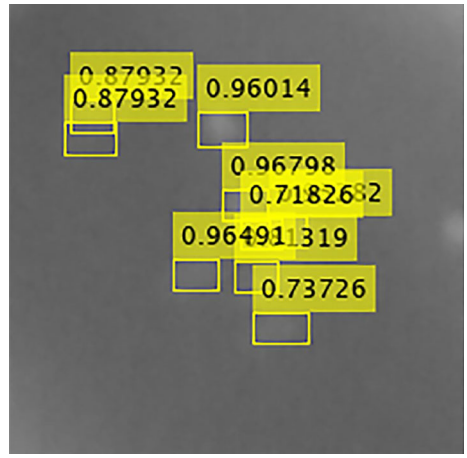


Fig. 8 Predicted image defect

The precision-recall curve is plotted in Fig. 7 To display the positive predictive value (PPV) versus the true positive rate (TPR) of the Faster RCNN detector.

Table 4 furnishes comprehensive insights into the prediction outcomes and associated annotation details for the images illustrated in Fig. 8. The scores presented in this image correspond to the prediction scores assigned to each bounding box, each of which identifies a detected defect. These scores are automatically generated as part of the prediction process. While the visual representation of these scores is provided in the image, a comprehensive listing and comparison of these results can be found in Table 4. The porosity box column displays the predicted bounding box encompassing the porosity, while the G porosity box column showcases the actual ground truth bounding box covering the porosity. The score column exhibits the predicted probability value, Prob label designates the anticipated weld failure type, and the true label column indicates the actual type of weld defect.

The results show that the model accurately locates the weld fault zone. The porosity defect in the picture is clear, but the model may have some errors in classifying the welding defects. In some cases, it may be easy to determine the type of defect, but accurately identifying the real coordinates of the defect can be a problem. This is an area that could be optimized in future work to improve the model's performance.

The test results indicate that the method of using a sliding window size of 7 × 7 achieves the best performance compared to other deep convolutional neural network (DCNN) networks used as feature extractors for the Faster RCNN model. This model outperforms other object detection models, such as the YOLOv2 model, which will be discussed further in the next section. The model shows superior effectiveness in detecting various types of porosity defects. When there are numerous defects in each image, it can result in overlapping bounding boxes. However, despite this challenge, the model still performs well in detecting and localizing the defects. One important factor con- tributing to the high performance of the trained model is the increase in data for different defect types. By having a large and diverse dataset, the model becomes more robust and capable of accurately identifying various types of defects. Overall, the outcomes demonstrate that the selected method for identifying defects is effective. The trained model demonstrates improved performance compared to other object detection models and is particularly effective in detecting various porosity defect

Table 4 Predicted detection for the image mentioned in Fig. 8

Porosity	Porosity box	G porosity box	Score	Prob label	True label
Object1	[96, 52, 23, 18]	[64,53, 14, 19]	0.9679798	Porosity	Porosity
Object2	[16, 22, 29, 42]	[17, 19, 21, 42]	0.9649069	Porosity	Porosity
Object3	[77, 125, 28, 16]	[73, 94, 12, 15]	0.9528173	Porosity	Porosity
Object4	[104, 91, 23,16]	[87, 108, 16, 16]	0.9601426	Porosity	Porosity
Object5	[115, 143, 26, 16]	[82, 135, 12, 14]	0.7182605	Porosity	Porosity
Object6	[115, 128, 24,16]	[53, 127, 15, 16]	0.8131949	Porosity	Porosity
Object7	[127, 153, 22, 16]	[80, 159, 18, 13]	0.8793176	Porosity	Porosity

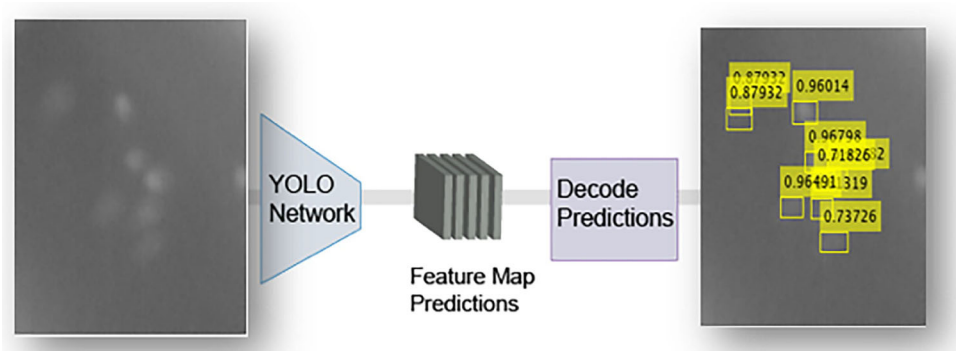


Fig. 9 The detailed visualization of the object detection model

types but further improvements in model optimization and data collection may be necessary to ameliorate the accuracy of defect recognition.

5.3 The Time of Processing

As mentioned previously, the Faster RCNN model achieves a processing time of 4 h when utilizing a GPU. This corresponds to handling 2 frames every second on a standard personal computer without the use of additional image processing hardware. In order to achieve real-time processing, we conducted a detection and classification operation every half-second. The accuracy process yielded an average of 95.15%. With the Geforce RTX2080 GPU, the average detection time per image is 2 s. Figure 8 illustrates sample testing results, highlighting that detection with higher confidence levels shows greater accuracy. Consequently, there is a notable rise in the count of true positives at high confidence levels.

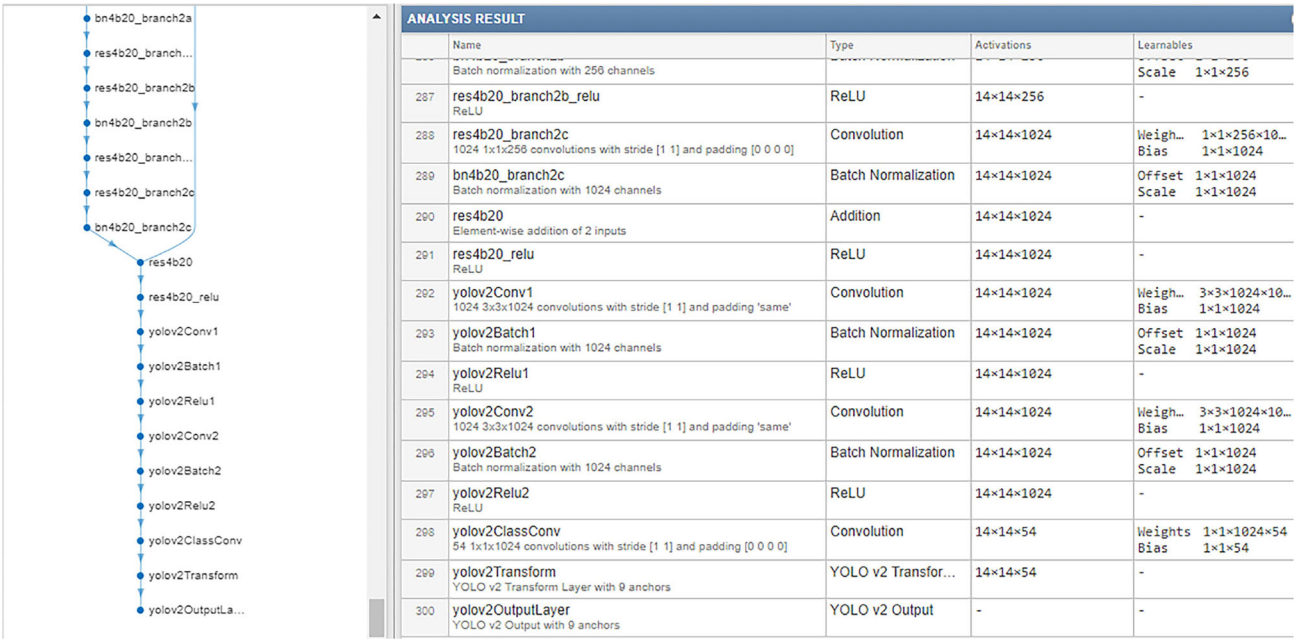
6 Comparative Analysis with YOLOv2

6.1 YOLOv2 Overview

The You Only Watch Once (YOLO) system introduces a groundbreaking approach to real-time object detection. YOLOv2 leverages a faster single-stage network in contrast to the two-stage object detectors like Faster R-CNN. In this study, we employ YOLOv2, which utilizes a CNN to perform deep learning on input images, generating accurate network predictions. The object detector then decodes these predictions, resulting in precise bounding boxes. The visual representation of the model architecture is shown in Fig. 9. The YOLOv2 object detection network consists of two interconnected subnetworks: the feature extraction and detection networks. The feature extraction network, commonly based on a pretrained CNN, plays a vital role in capturing meaningful features. In our study, we employ ResNet101 as the pretrained CNN for feature extraction. On the other hand, the detection subnetwork, although smaller compared to the feature extraction network, is crucial for precise object detection. It comprises several convolutional layers and layers specifically designed for YOLOv2.

To facilitate the creation of a YOLOv2 object detection network, we lever- age the yolov2Layers Computer Vision function. This powerful tool automatically constructs a YOLOv2 network using a pre-trained ResNet101 feature extraction network as the base. By eliminating all layers that come after the layer feature located in the main network and seamlessly integrating the detection sub-network, Yolov2Layers streamlines the model development process. The detection sub-network comprises a series of connected convolution, ReLU, and batch normalization layers. Additionally, the YOLOv2 transform layer and YOLOv2 output layer are added to the detection sub-network. The transform layer is responsible for converting the raw CNN output into a suitable format for object detection, while the output layer defines the anchor box paramters and implements the loss function utilized during detector training. The detailed visualization of the object detection model's design is shown in Fig. 10.

To set up a YOLOv2 network with Yolov2Layers, you must define various parameters, including the network's input size, anchor boxes, and the feature extraction network. The input size is particularly significant as it not only establishes the network's minimum size requirement but also aligns with the training image dimensions and influences computational demands. It's recommended to select an input size that closely matches the training image dimensions while meeting the network's minimum criteria. In this instance, the input size is configured as [224 224 3]. Next, the anchor boxes are estimated based on the object sizes observed in the training data. To achieve this, the Anchor Boxes Estimates Toolbox is employed, which analyzes the training data and provides an estimation of the anchor boxes. These anchor boxes serve as reference points for object detection during the network's operation. Before proceeding further, the training data undergoes processing through a transformation function to ensure compatibility with the network. Subsequently, the



	Name	Type	Activations	Learnables
	Batch normalization with 256 channels			Scale 1×1×256
287	res4b20_branch2b_relu ReLU	ReLU	14×14×256	-
288	res4b20_branch2c 1024 1x1x256 convolutions with stride [1 1] and padding [0 0 0 0]	Convolution	14×14×1024	Weigh… 1×1×256×10… Bias 1×1×1024
289	bn4b20_branch2c Batch normalization with 1024 channels	Batch Normalization	14×14×1024	Offset 1×1×1024 Scale 1×1×1024
290	res4b20 Element-wise addition of 2 inputs	Addition	14×14×1024	-
291	res4b20_relu ReLU	ReLU	14×14×1024	-
292	yolov2Conv1 1024 3x3x1024 convolutions with stride [1 1] and padding 'same'	Convolution	14×14×1024	Weigh… 3×3×1024×10… Bias 1×1×1024
293	yolov2Batch1 Batch normalization with 1024 channels	Batch Normalization	14×14×1024	Offset 1×1×1024 Scale 1×1×1024
294	yolov2Relu1 ReLU	ReLU	14×14×1024	-
295	yolov2Conv2 1024 3x3x1024 convolutions with stride [1 1] and padding 'same'	Convolution	14×14×1024	Weigh… 3×3×1024×10… Bias 1×1×1024
296	yolov2Batch2 Batch normalization with 1024 channels	Batch Normalization	14×14×1024	Offset 1×1×1024 Scale 1×1×1024
297	yolov2Relu2 ReLU	ReLU	14×14×1024	-
298	yolov2ClassConv 54 1x1x1024 convolutions with stride [1 1] and padding [0 0 0 0]	Convolution	14×14×54	Weights 1×1×1024×54 Bias 1×1×54
299	yolov2Transform YOLO v2 Transform Layer with 9 anchors	YOLO v2 Transfor...	14×14×54	-
300	yolov2OutputLayer YOLO v2 Output with 9 anchors	YOLO v2 Output	-	-

Fig. 10 YOLOv2 model Lgraph

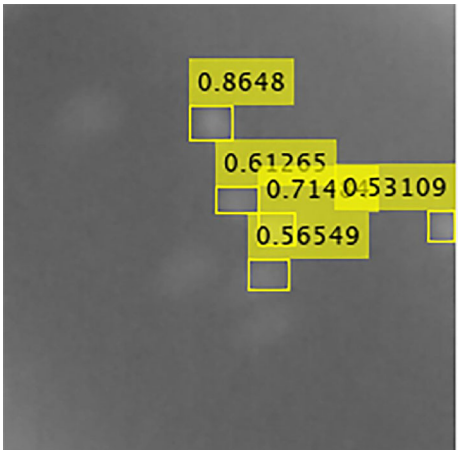


Fig. 11 Result of defect detection

determination of the anchor boxes number is contingent upon the unique demands of the application, and their estimated values are established through suitable techniques.

6.2 Experimental Results of YOLOv2

Our algorithm has been implemented using MATLAB, providing us with a powerful platform for developing and training deep learning models. With the support of MathWorks, we utilize the deep learning framework to construct and train our models, as well as make predictions using these trained models. To evaluate the performance of our approach, we conducted extensive testing, and the results are depicted in Fig. 11. The scores presented in this image correspond to the prediction scores assigned to each bounding box, each of which identifies a detected defect. These scores are automatically generated as part of the prediction process. While the visual representation of these scores is provided in the image, a comprehensive listing of these results can be found in Table 5. The score column exhibits the predicted probability value, Prob label designates the anticipated weld failure type, and the True label column indicates the actual type of weld defect.

Table 5 Predicted detection for the image mentioned in Fig. 11

Porosity	Score	Prob label	True label
Object1	0.8648	Porosity	Porosity
Object2	0.61265	Porosity	Porosity
Object3	0.71484	Porosity	Porosity
Object4	0.56549	Porosity	Porosity
Object5	0.53109	Porosity	Porosity

We also considered the inference time of each model and assessed various metrics to compare the performance of Faster RCNN and YOLOv2, as presented in Table 6. The comparative analysis demonstrates that our model outperforms the latter design. Additionally, the identification of predicted defects in the test image (Fig. 8) validates the robustness of our detection model.

YOLO exhibits certain limitations compared to the Faster RCNN model. It tends to have more location errors, and its

Table 6 Models performance

AP (%)	Mean RMSE	Mean loss	Anchors number	Training time (h)
YOLOv2 + aug 39%	0.41	0.34	9	4:05
Faster RCNN 44.5%	0.39	0.33	9	4

recall rate is relatively lower when compared to region proposal methods. So, our primary emphasis lies in enhancing recall and location accuracy while upholding classification accuracy. Achieving exceptional performance often involves the utilization of larger net- work architectures or the fusion of multiple models. In essence, our findings underscore the effectiveness of our approach when compared to other DCNN networks and the YOLOv2 model.

7 Conclusions

In this study, we have employed the advanced deep learning technique, Faster- RCNN, to effectively address the challenge of weld defect detection in X-ray images. Unlike conventional approaches, this algorithm significantly enhances detection capability by operating in a high-dimensional domain. By accurately identifying and localizing the welding defect regions, along with providing the corresponding probability of defect types, the system empowers inspectors to efficiently examine X-ray images. To ensure optimal training performance, we have emphasized the importance of image enhancement and noise reduction techniques. The experimental results demonstrate the effectiveness of our proposed object detection system, achieving a satisfactory average precision (AP) of 44.5%. This performance showcases promising potential in overcoming the limitations of traditional methods, particularly in handling diverse object sizes and addressing data scarcity challenges.

Our main challenge is to solve a real industrial problem mainly related to the visual inspection of the X-ray welding database not to invest in an available benchmark. The lack of X-ray welding images is a challenge that needs to be adjusted especially if they are acquired with low quality and a variety of types. Moreover, the detection of X-ray weld defects with the abundance of small sticky porosity in every single image leads to several issues related to the selection of the proper deep neural network object detector and its performance.

In response to the limited availability of radiographic testing data, we incorporated data augmentation techniques to facilitate efficient training and improve overall performance. The findings demonstrate that this approach significantly enhances the testing image performance. In response to the scarcity and diversity of X-ray welding images, a novel approach has been developed to detect weld defects using an object detection model giving particular attention to the detection of small-sized defects called"Porosity". This defect detection process necessitates adjustments to the anchor box size and aspect ratio to better suit these objects. Through experimentation, the optimal number of region proposal recommendations was determined. Our contribution is a continuity of my previous publication [38] which has focused on the weld classification of welding defects. So, the succession of these two computer vision tasks could be implemented in the future to automate the visual inspection process.

Comparisons of various deep convolutional neural network (DCNN) models highlighted the superior performance of our proposed ResNet model, even outperforming YOLOv2. In summary, this work underscores the efficiency of applying a pre-trained model for weld defects classification, followed by object detection, for the visual inspection of X-ray data.

Looking ahead, this study presents a comprehensive approach to weld defect detection that not only addresses current challenges but also sets the stage for future advancements in the realm of automated visual inspection processes within the field of X-ray data analysis. Our future plans involve dataset augmentation, algorithm refinement to address welding defects more specifically, expanding the application scope, and exploring semi-supervised learning techniques. Additionally, the prospect of implementing a comprehensive pipeline combining segmentation and object detection techniques for a more robust and accurate defect detection system holds great promise.

Acknowledgements This work has been partially funded by the Spanish Government through Project RTI2018-097088-B-C33 (MINECO/FEDER, UE).

Author contributions The primary contributor to this manuscript, including the writing, results, and innovation, is the corresponding author. The remaining authors were involved in a minor manuscript revision.

Funding It is not applicable.

Data Availability It is not applicable.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval It is not applicable.

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